Extracellular matrix and glycoconjugates

The **extracellular matrix** (ECM), literally "any material outside the cell", is a complex three-dimensional molecular network that surrounds cells and fills the space between them. Connective tissue is especially rich in extracellular matrix and provides structural and mechanical support throughout the body. The composition and organization of the extracellular matrix vary considerably among tissues, giving rise to very different physical properties. For example, bone contains a mineralized, rigid matrix; tendons contain collagen-rich matrices with high tensile strength; cartilage resists compressive forces; the cornea has a highly organized, hydrated and transparent matrix; and basement membranes form specialized matrix structures that support and organize epithelial and other cell layers.

In addition to providing structural support, the extracellular matrix is a dynamic and physiologically active environment that regulates cell adhesion, migration, proliferation, differentiation, and responses to extracellular signals. The composition and organization of the extracellular matrix largely determine the structure and function of our tissues.

Major components of the extracellular matrix include **structural fibrous proteins**, adhesive glycoproteins, proteoglycans and glycosaminoglycans. Many of these components are **glycoconjugates**, molecules in which carbohydrates are covalently linked to proteins or lipids. Glycoconjugates include proteoglycans, glycoproteins and glycolipids. They are not restricted to the extracellular matrix: they are also abundant on the cell surface and in secreted molecules.

Several extracellular matrix proteins, including collagen, fibronectin and laminin, bind to **integrins**, transmembrane receptors that mediate cell–matrix adhesion and signaling. Upon binding, integrins can modulate signal transduction pathways that mediate cellular processes such as the cell cycle, the organization of the intracellular cytoskeleton, or the trafficking of cell membrane receptors.

Matrix metalloproteinases (MMPs) are enzymes that catalyze the cleavage of proteins found in the extracellular matrix. MMPs continuously remodel the extracellular matrix, regulating cell migration and responses to extracellular signals. The activity of MMPs is regulated by tissue inhibitors of metalloproteinases (TIMPs).

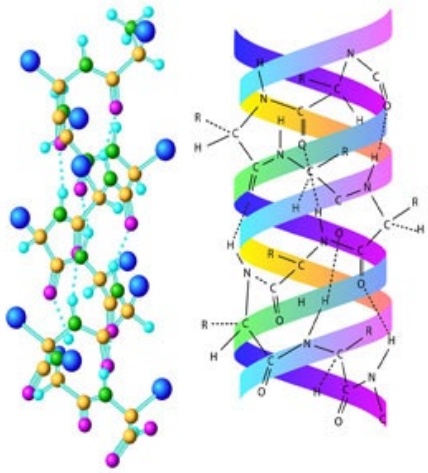
Fibrous proteins

Fibrous proteins are polypeptide chains characterized by elongated structures formed by repeating structural elements. Collagen and elastin are major fibrous proteins of the extracellular matrix. Other fibrous proteins, such as α-keratin, are intracellular proteins.

α-Keratin

α-Keratins are characterized by their **strength**. In humans, they constitute the main component of the epidermis, hair and nails (and, in other mammals, horns, claws, etc.). They are not proteins of the extracellular matrix, but they accumulate inside the cells, especially as these cells mature and die to form protective outer layers.

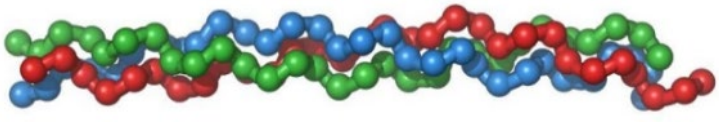
α-Keratins are composed of **two right-handed α helices wrapped around each other in a left-handed sense** to form a supertwisted coiled coil. The surfaces where the two helices are in contact are made up of hydrophobic amino acids. This distribution permits a close packing of the polypeptide chains. α-Keratins are **rich in hydrophobic residues**, such as Ala, Val, Leu, Ile, Met and Phe. The supertwisted dimer structures can be assembled into large complexes forming fibers cross-linked by disulfide bonds.



Collagen

Collagens make up strong fibers that provide **tensile strength** to tissues. **Collagen is the most abundant protein in the human body**. Collagen is the principal constituent of cartilages, bones, tendons, ligaments, skin and blood vessels.

Collagen is a **right-handed triple helix composed of three left-handed helices**. These latter helices contain approximately 1,000 amino acid residues and are made up of a repeating tripeptide unit: **Gly-X-Y**, where X is often **proline** and Y is often **4-hydroxyproline**. The three polypeptide chains are linked by hydrogen bonds. Each turn of the triple helix contains three amino acid residues, such that every third amino acid is in close contact with the other two strands in the center of the complex. Only **glycine** can fit in this position. Fibrillar collagens assemble into collagen fibrils stabilized by covalent bonds.



Collagen is synthesized as prepro-α chains. After removal of the signal peptide in the endoplasmic reticulum, the resulting pro-α chains undergo extensive posttranslational modifications. Three modified pro-α chains then assemble into the triple-helical procollagen molecule. Procollagen is secreted from the cell, and its N- and C-terminal propeptides are cleaved to form collagen (tropocollagen) molecules, which assemble into fibrils and are subsequently stabilized by covalent cross-links.

Posttranslational modifications include the **hydroxylation** of proline and lysine residues to produce **hydroxyproline** and **hydroxylysine** residues. These reactions require **ascorbic acid** (vitamin C) as a cofactor of the enzymes *prolyl hydroxylase* and *lysyl hydroxylase*. Hydroxyproline

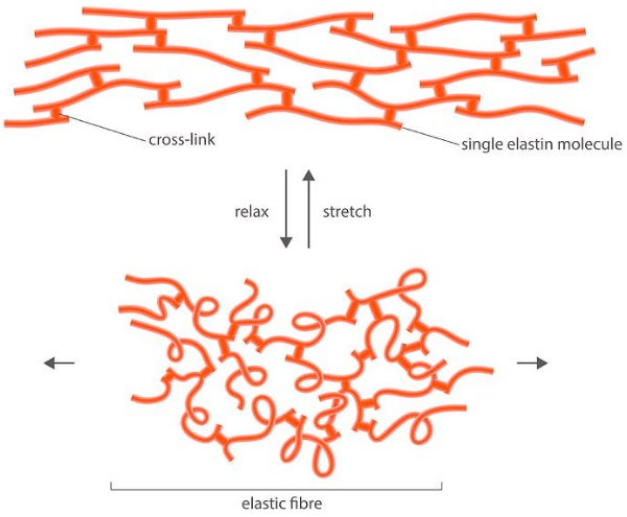
forces the proline ring into a more favorable conformation and is involved in hydrogen bond formation helping to stabilize the triple helix. Some hydroxylysine residues are glycosylated and also participate in collagen cross-linking. Another modified residue that participates in the formation of covalent bonds is allysine (from the oxidation of lysine in a reaction catalyzed by the enzyme *lysyl oxidase*).

Twenty-eight types of collagen have been identified in humans. Each tissue has a particular composition of collagens, which contributes to its specific mechanical properties. Collagens can be classified into fibril-forming, network-forming, and other types, including some transmembrane collagens. Approximately 90% of the collagen in the human body is **type I collagen**. The triple helix of type I collagen has higher tensile strength than a steel wire of equal cross section.

Elastin

Elastin is the major protein found in elastic fibers of the extracellular matrix. Elastin permits tissues to return to their original shape after they have been stretched. This is of special importance to blood vessels, which expand and contract repeatedly in response to changes in intravascular pressure. Elastin is rich in **hydrophobic amino acids**, such as alanine, valine, glycine and proline.

Elastin is synthesized as a water-soluble precursor named **tropoelastin**. Upon secretion from the cell, tropoelastin forms fibrils **crosslinked through lysine residues** to generate an elastic network that can be stretched and subsequently recoil to its original dimensions.



Source: Science Photo Library

Adhesive glycoproteins

Fibronectin

Fibronectin serves several functions. Fibronectin helps position cells within the extracellular matrix. Some cells use the fibronectin matrix to migrate, which is particularly important during embryonic development and wound healing. Fibronectin can also regulate proliferation.

Fibronectin is a dimer consisting of two nearly identical polypeptide chains linked by two C-terminal disulfide bonds. It is secreted in a soluble form and subsequently assembles into an insoluble fibrous mesh. Each fibronectin subunit consists of repeating type I, type II and type III modules, which form functional domains that bind integrins and other extracellular matrix components, such as collagen. The **RGD (Arg-Gly-Asp) sequence** in fibronectin is a key cell-binding motif recognized by several integrins, allowing cells to attach to the extracellular matrix.

Laminin

Laminin is a **heterotrimeric protein** (α, β and γ subunits). The α, β and γ chains assemble into a heterotrimer stabilized by disulfide bonds. After secretion, laminin molecules polymerize to form extracellular networks and interact with integrins and other basement membrane components. **It is a major glycoprotein of basement membranes**, together with collagen IV. Laminin is synthesized as a precursor and, upon secretion, is assembled into extracellular networks stabilized by disulfide bonds.

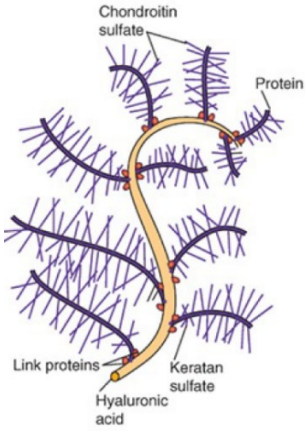
Glycoconjugates

Glycoconjugates are molecules in which carbohydrates are covalently linked to proteins or lipids. There are three major types of glycoconjugates: proteoglycans, glycoproteins and glycolipids. Proteoglycans and some glycoproteins are important components of the extracellular matrix, whereas glycoproteins and glycolipids are also abundant at the cell surface, where they participate in cell recognition, adhesion and signaling.

Proteoglycans

A **proteoglycan** is a macromolecule composed of a **core protein covalently linked to long glycosaminoglycan chains**. Carbohydrates may account for most of the mass of a proteoglycan. With the important exception of hyaluronan, glycosaminoglycan chains are generally covalently attached to core proteins to form proteoglycans. Whereas collagen provides tensile strength, proteoglycans provide resistance to compressive forces. This property stems primarily from the glycosaminoglycan portion of proteoglycans. They also affect other cellular activities, such as growth and adhesion. Proteoglycans interact with a variety of fibrous proteins in the matrix, such as collagen or fibronectin. They can also form **proteoglycan aggregates**, supramolecular assemblies of many proteoglycans bound to a chain of hyaluronan.

There are several types of proteoglycans that differ in the glycosaminoglycan chains. Many are free in the extracellular matrix but some are membrane-associated proteins. Proteoglycans are major components of connective tissues such as cartilage. Aggrecan, the major proteoglycan of cartilage, is rich in chondroitin sulfate and keratan sulfate. The complementary functions of proteoglycan and collagen molecules give cartilage its resilience (cartilage can withstand a high compressive load and then reexpand to its previous dimensions when that load is relieved).



Source: Mark's Basic Medical Biochemistry. Ed. Wolters Kluwer

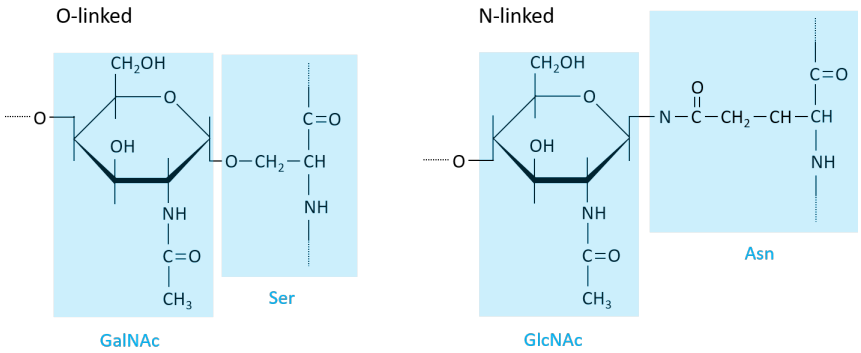
Examples of proteoglycans anchored to membranes are the members of the syndecan and the glypican families, which act as membrane co-receptors. **Syndecans** have a protein transmembrane domain and an extracellular domain bearing some glycosaminoglycan chains. **Glypicans** are attached to the membrane by a lipid anchor.

Glycoproteins

Glycoproteins are **proteins covalently linked to carbohydrates**. Normally, these carbohydrates are small (oligosaccharides), branched and structurally diverse, both in terms of monosaccharide types and glycosidic bonds. The structural diversity is in clear contrast with that of polysaccharides (highly homogeneous polymers) and determines that each oligosaccharide has its own peculiarities in terms of its interactions (with enzymes, receptors …). The carbohydrate content in glycoproteins ranges from 1 to 70% of the total weight of the glycoprotein.

Oligosaccharide chains can bind to the protein by two different linkages:

- **O-glycosidic bond**: the anomeric carbon of the first residue in the oligosaccharide, typically **N-acetylgalactosamine**, binds to the hydroxyl group of **serine** or **threonine** residues in the protein.
- **N-glycosidic bond**: the anomeric carbon of the first residue in the oligosaccharide, normally **N-acetylglucosamine**, binds to the amide of an **asparagine** residue.



N-linked glycosylation begins in the endoplasmic reticulum during protein synthesis and continues to be processed during transit through the Golgi apparatus. In contrast, most O-linked glycosylation begins in the Golgi.

More than half of human proteins are glycoproteins. Most secreted and membrane proteins are glycosylated. Plasma glycoproteins commonly contain N-linked glycans, and some also contain O-linked glycans (albumin is a notable example of a nonglycosylated plasma protein). The external surface of the plasma membrane has many membrane glycoproteins. **Mucins** are secreted or membrane-bound glycoproteins that contain large numbers of O-linked oligosaccharides. Mucins are present in most mucous secretions. They are found in abundant amounts in salivary secretions and in the epithelium, where they act as a lubricant and protective barrier.

The oligosaccharide chains of glycoproteins can perform a variety of functions. Some general functions of oligosaccharides of glycoproteins are:

1- Steric interactions between the polypeptide chain and the oligosaccharide can influence the folding of the protein.

2- The high hydrophilicity of the carbohydrate chain can alter the polarity and solubility of the protein.

3- The presence of an oligosaccharide in a protein can establish a recognition site for other molecules (the oligosaccharide “marks” the protein). This can allow or prevent the binding to enzymes, receptors or cells.

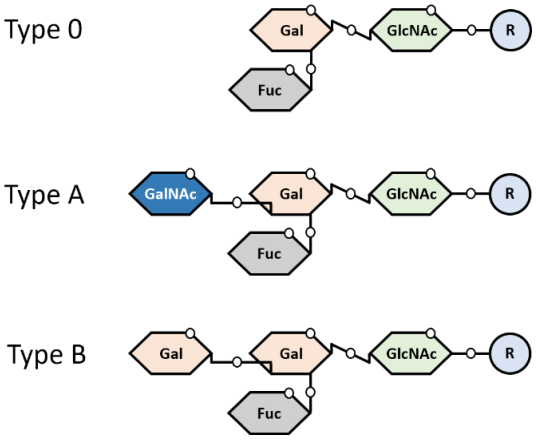
Lectins are proteins that bind carbohydrates with high specificity and affinity. Lectins serve in a wide variety of cell-cell recognition, signaling and adhesion processes. Some lectins are mannose-6-phosphate receptors that recognize mannose-6-phosphate residues in newly synthesized lysosomal enzymes and direct their trafficking toward endosomes and lysosomes. **Selectins**, a family of plasma membrane lectins, mediate leukocyte adhesion and rolling along the vascular endothelium, an important step in leukocyte extravasation at sites of infection or inflammation. Several pathogens attach to their host cells through their glycoproteins or their lectins (Influenza virus, *Helicobacter pylori*, *Bordetella pertussis* or HIV, among many others).

Glycolipids

Most mammalian **glycolipids** are glycosphingolipids derived from **sphingosine**. Glycolipids are in the plasma membrane, where they can have cell recognition functions. Two important types of glycolipids are cerebrosides and gangliosides. **Cerebrosides** consist of a ceramide with a single

sugar residue (glucose or galactose). In **gangliosides**, the ceramide is bound to an oligosaccharide chain. As in glycoproteins, the oligosaccharide chains of gangliosides are highly diverse in composition and structure.

The **ABO blood group antigens** are oligosaccharide structures present on cell-surface glycoproteins and glycolipids. Depending on the type of *glycosyltransferases* present in an individual, glycoconjugates contain different oligosaccharides. Blood group O retains the unmodified H antigen, whereas group A and group B *glycosyltransferases* add terminal N-acetylgalactosamine and galactose residues, respectively. Antibodies to A and to B antigens occur naturally in the blood of people whose red blood cells lack the corresponding antigen. These antibodies may cause intravascular destruction of red blood cells if given inadvertently during a blood transfusion.



Adapted from: Mark's Basic Medical Biochemistry. Ed. Wolters Kluwer

Lipopolysaccharides consist of lipid A, a phosphorylated glucosamine disaccharide bearing several fatty acid chains, linked to a core oligosaccharide and a variable O-antigen polysaccharide. Lipopolysaccharide is the main component of the outer membrane of Gram-negative bacteria (e.g. *E. coli* and *Salmonella*). Lipopolysaccharides are also known as **endotoxins** because lipid A strongly activates innate immune responses and, when present systemically at high levels, can contribute to septic shock.

Pathologies associated with impaired glycoconjugate degradation

Lysosomal glycosidases degrade proteoglycans, glycoproteins and glycolipids, which are brought into the cell by endocytosis. Defects in lysosomal enzymes responsible for glycosaminoglycan degradation cause **mucopolysaccharidoses**, a group of lysosomal storage diseases characterized by the accumulation of partially degraded glycosaminoglycans within lysosomes. This accumulation causes lysosomal enlargement and impairs cellular and tissue function. Mucopolysaccharidoses often affect skeletal development and multiple organs, and some types also involve the nervous system.

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